Lab 3 Superposition, Thevenin and Norton Equivalents, and Maximum Power Transfer

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Objectives

The purpose of this experiment is to verify experimentally the superposition, Thevenin and Norton Theorems, and to investigate the maximum power transfer condition.

Equipment and Components

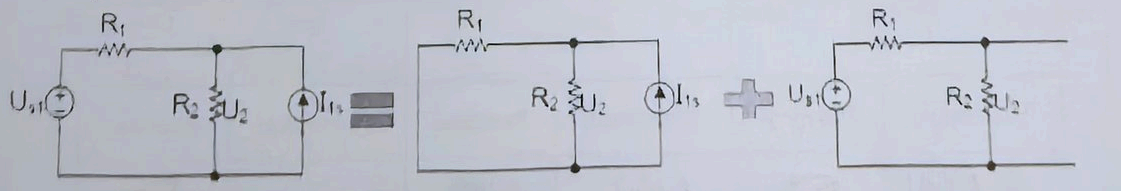
- DC Power Supply
- Digital Multimeter (DMM)
- Current-source board
- Breadboard, jumper wires and test cables
- Resistors specified in the circuit diagrams
- Potentiometer W102 for adjusting the equivalent resistance
- Potentiometer W103 for adjusting the load resistance

Experiment 1: Superposition Theorem

Superposition theorem states that, in a linear circuit containing multiple independent sources, the voltage across or current through an element is the algebraic sum of the contributions produced by each independent source acting alone. When evaluating one source contribution, set all other independent sources to zero. Dependent sources, if present, must remain active. Superposition applies to voltage and current, but not directly to power, because power is a nonlinear function of voltage or current.

Source Deactivation

- For a **nonideal** source modeled as an ideal source with an internal resistance, set the ideal source to zero while retaining its internal resistance.
- If the sources are **ideal**, in theory:
  Replace an ideal independent voltage source with a short circuit.
  Replace an ideal independent current source with an open circuit.



(a) Both sources active alone (b) Current source acting alone (c) voltage source acting

Figure 1. Illustration of superposition.

Unless otherwise specified, treat the independent voltage and current sources as ideal in the theoretical calculations.

Verification of Superposition Theorem in a DC Circuit.

Consider the circuit given in Figure 2.

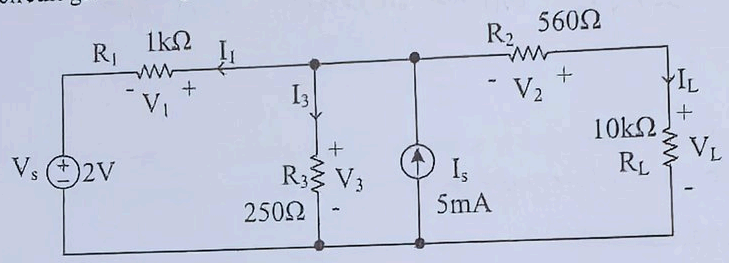


Figure 2. Circuit used to verify superposition.

1. Pre-Laboratory Calculations

Consider the circuit shown in Figure 2. Using the nominal component values, calculate the load voltage V_L and load current I_L for each of the following conditions:

1) With both independent sources active : V_L and I_L.
2) The current source I_S acting alone, with the voltage source V_S replaced by a short circuit: V_L' and I_L'.
3) The voltage source V_S acting alone, with the current source I_S replaced by an open circuit: V_L'' and I_L''.

Use the same voltage polarity and current reference direction for all three conditions. Record the signed calculated values in Table 1. (12 pts)

Table 1. Calculated and Measured Results for Superposition

V_L (V)		V_L' (V)		V_L'' (V)	
Theoretical	Practical	Theoretical	Practical	Theoretical	Practical
1.3	1.305	0.93	0.939	0.37	0.37
I_L (mA)		I_L' (mA)		I_L'' (mA)	
Theoretical	Practical	Theoretical	Practical	Theoretical	Practical
0.13	0.12	0.093	0.09	0.037	0.03

2. Measure the values of the resistors using the multimeter before installing them in the circuit. If a resistor is already installed, disconnect at least one terminal before measuring its resistance. Record the values in Table 2. (4 pts)

Table 2. Nominal and Measured Resistance Values

Resistor	Nominal Value	Ohmmeter Reading
R_1	1 kΩ	998Ω.
R_2	560 Ω	559 Ω.
R_3	250 Ω	299.6Ω.
R_L	10 kΩ	9.98kΩ.

3. After measuring the actual resistor values, construct the circuit on your breadboard according to the schematic shown in Figure 2. Verify all wiring connections, source polarities, and the current-source direction before applying power.

1) **Both sources active**: Measure the load voltage V_L across the load resistor R_L and the current I_L through R_L. Connect the **voltmeter in parallel with** R_L and the **ammeter in series with** R_L.

4) **Current source I_S acting alone** (Deactivate the voltage source V_S): Turn off the power before changing any connections. Disconnect the DC voltage supply from the circuit, then connect a jumper wire between the two circuit nodes previously connected to the voltage source. Apply power to the current-source board and measure V_L' and I_L'.

5) **Voltage source V_S acting alone** (Deactivate the current source I_S): Turn off the power before changing any connections. Replace the current-source branch with an open circuit. Restore the voltage-source connection and set V_S to the value specified in Figure 2. Apply power and measure V_L'' and I_L''.

2) Record all measured load voltages and currents in Table 1.

$$Percentage\ Error = \frac{|Experiment\ Value - Theory\ Value|}{Theory} \times 100\%$$

4. Compare the measured and calculated load voltages and currents for each source condition. (3 pts)

Measured load voltages and currents agree closely with the calculated values.

5. Verify superposition by comparing the directly measured values with the algebraic sums. (4 pts)

0.939 + 0.370 = 1.305 V compare with 1.305 V

0.09 + 0.03 = 0.12 mA = 0.12 mA

Experiment 2: Thevenin's Theorem and Equivalent Parameters

Thevenin and Norton equivalents are circuit simplification techniques that focus on terminal behavior and thus are extremely valuable aids in analysis. Figures 3(a) and 3(b) show the original network and its Thevenin equivalent. When connected to the same load, the two circuits produce the same load voltage and load current.

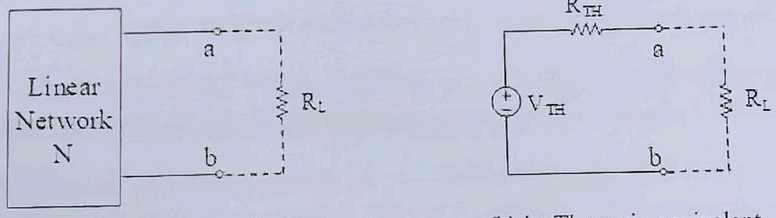


Figure 3. (a) Original linear two-terminal network (b) its Thevenin equivalent.

1. Thevenin Parameters

1) Thevenin Voltage V_{TH} : The Thevenin voltage is the open-circuit voltage across the terminals **a - b**:

$$V_{TH} = V_{OC}$$

2) For a network containing only resistors and independent sources, Thevenin equivalent resistance R_{TH} is the resistance looking into terminals a-b of the original given circuit after all independent sources have been deactivated.

2. Methods for Determining the Thevenin Resistance

The following three methods can be used to determine the Thevenin resistance of the resistive network used in this experiment.

1) Open-circuit Voltage and Short-Circuit Current Method

Firstly, disconnect the load and use the DMM in voltage mode to measure the open-circuit voltage V_{OC} across terminals a–b.

Secondly, if short-circuit operation is safe, use the DMM in current mode to measure the short-circuit current I_{SC}, defined as flowing externally from terminal a to terminal b.

Finally , calculate the equivalent resistance of the two-terminal network: $R_{TH} = \frac{V_{OC}}{I_{SC}}$.

If the internal resistance of the active two-terminal network is very small, it is not suitable to use this method, because if the output port is short-circuited, it is easy to damage its internal components.

2) Voltage-Current Method

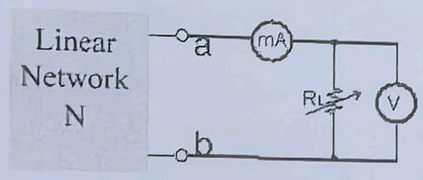


Figure 4. Measurement setup for the voltage–current method.

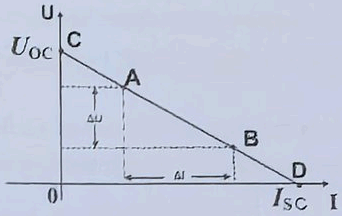


Figure 5. External characteristic of the linear two-terminal network.

Attach a potentiometer across the output terminals **a** and **b**, that is the output end of the active two-terminal network is loaded with a variable resistor R_L. Use a voltmeter and an ammeter to measure the load voltage V_L and the current I_L delivered to the load.

Plot V_L versus I_L as shown in Figure 5. The voltage-current relationship measured when the resistor R_L is given different values is called the output characteristic or external characteristic of the active two-terminal network.

The slope of the measured voltage–current characteristic can be used to determine the Thevenin resistance:

$$R_{TH} = \frac{\Delta V}{\Delta I} = \frac{V_{oc}}{I_{SC}}$$

For the Point C, $I = 0$, $V = V_{oc}$; For the Point D, $V = 0$, $I = I_{SC}$.

3) Half-Voltage Method

As we know, if two resistors making up a divider have the same resistance, the voltage across either one will be exactly half of the source voltage. As a result,

- Measure the open-circuit voltage V_{OC}
- Attach a potentiometer across terminals a-b, as shown in Figure 6.
- Adjust the potentiometer until the output voltage is exactly half of the open-circuit voltage:

$$V_L = \frac{V_{OC}}{2}.$$

When $V_L = \frac{V_{OC}}{2}$, the potentiometer resistance is equal to the Thevenin resistance.

- Then turn off the power, disconnect the potentiometer from the circuit without changing its setting, and measure its resistance using the DMM.

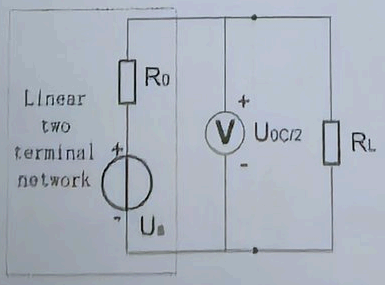


Figure 6. Measurement setup for the half-voltage method.

3. Norton's Theorem

Norton's theorem states that a linear two-terminal circuit can be replaced by an equivalent circuit consisting of a current source I_N in parallel with a resistor R_N. The Norton parameters are

$$I_N = I_{SC}\,,\quad R_N = R_{TH}\,,$$

where I_N is the short-circuit current I_{sc} flowing from terminal a to terminal b, and R_N is the input or equivalent resistance at the terminals when the independent sources are turned off.

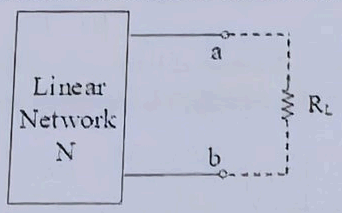


Figure 7(a) Linear two terminal network

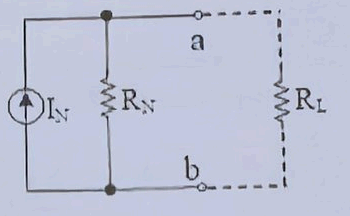


Figure 7(b) Norton equivalent circuit

Experimental Verification of Equivalent Circuits

The original network and an equivalent circuit should have the same terminal voltage–current relationship when connected to identical loads. Equivalence can be tested by the following steps:

1) Determine the Thevenin and Norton equivalent-circuit parameters.
2) Connect the potentiometer (load variable resistor) to the original circuit and measure the load voltage and current of the original network for a set of load resistances.
3) Construct the Thevenin/Norton equivalent circuit and repeat the measurements using the same load resistances.
4) Compare the two sets of results within experimental uncertainty.

In this experiment, perform these steps for the Thevenin equivalent. Design a corresponding procedure for the Norton equivalent in the discussion section.

Procedure

1. Turn off the power and remove R_L from terminals a-b of the circuit shown in Figure 2. Leave the rest of the original circuit intact. The remaining network is shown in Figure 8, with the output port open-circuited.

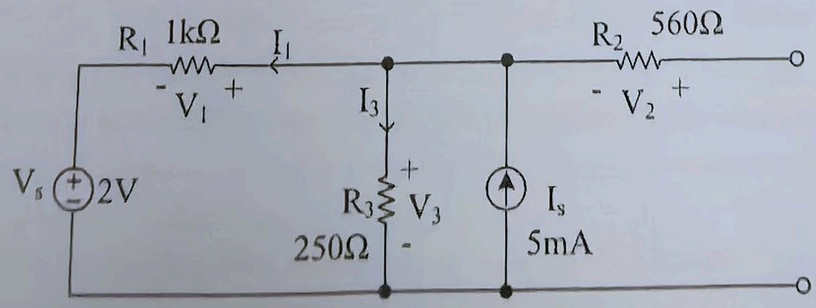


Figure 8

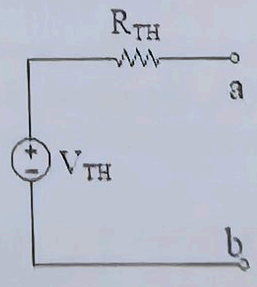


Figure 9

1) Measure the open-circuit voltage $V_{TH}=V_{OC}$ of the circuit shown in Figure 8 across terminals a–b.

2) Configure the DMM for current measurement and connect it across terminals a–b to measure the short-circuit current I_{SC}. In this manner, the ammeter will act as a short circuit.

3) Calculate the Thevenin equivalent resistance R_{TH} using the two measured values above.

$$R_{TH} = V_{OC}/I_{SC}$$

Record the results in Table 3. (9 pts)

Table 3. Measured and Calculated Equivalent-Circuit Parameters

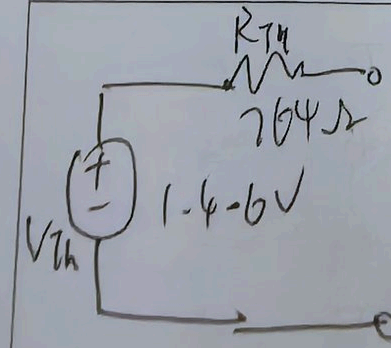
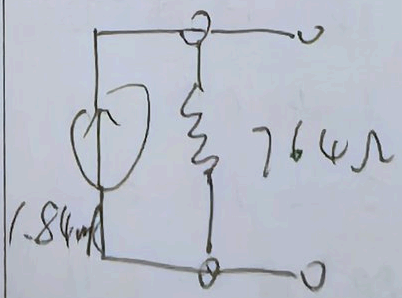
	Thevenin's Voltage V_{TH}	Short-circuit Current I_{SC}	Thevenin's equivalent Resistance $R_{TH} = V_{OC}/I_{SC}$
Theory Value	1.400 V	1.842 mA	760 Ω
Experiment Value	1.406V	1.84 mA.	764.
%Error	0.42 %	0.11 %	0.53%

Disconnect V_s from the circuit and replace it with a jumper wire (short circuit). This is equivalent to Vs = 0. Disconnect I_s from the circuit. With power off and the independent sources appropriately replaced, measure the resistance looking into terminals a–b, record this value as $R_{TH,ohm}$ and compare it with the value $R_{TH} = V_{OC}/I_{SC}$. (4 pts)

Vs disconnect: $V_{TH} = 0.481$ V. $I_{SC} =$ 1.28 mA.
Is disconnect: $V_{TH} = 0.398$ V. $I_{SC} = 0.51$ mA.

2. Verification of Thevenin's Theorem

1) Draw the Thevenin and Norton equivalent circuit using the values of V_{TH}, R_{TH} and I_{SC} recorded in Table 3. Clearly label the source magnitudes, voltage polarity or current direction, and resistance. (8 pts)

Thevenin Equivalent Circuit	Norton Equivalent Circuit
	

2) Leave the original circuit intact and construct the Thevenin equivalent circuit on the other side of the same breadboard using a DC power supply in series with a W102 potentiometer, as shown in Figure 9. Set the power supply voltage to the measured V_{TH} and adjust the potentiometer resistance to the measured R_{TH}.

3)

- Connect the load R_L (W103 potentiometer) across terminals a-b as shown in Figure 10. For each load resistance listed in Table 4, measure load voltage V_L and load current. I_L.
- Connect the load to the Thevenin equivalent circuit, as shown in Figure 11. Repeat the measurements using the same load resistance values.
- Repeat the measurements using the same load resistance values.
- Record the load resistance, voltage, and current for both circuits in Table 4. Turn off the power before changing connections. （28 pts）

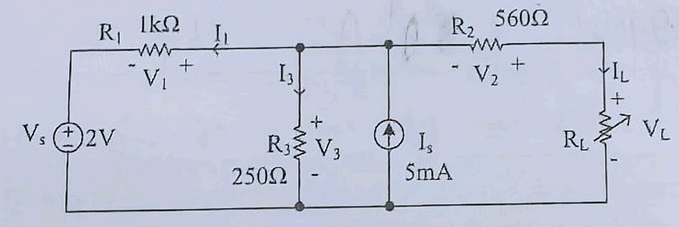


Figure 10

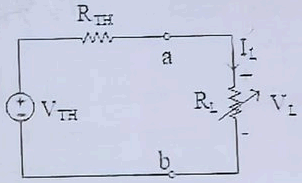


Figure 11

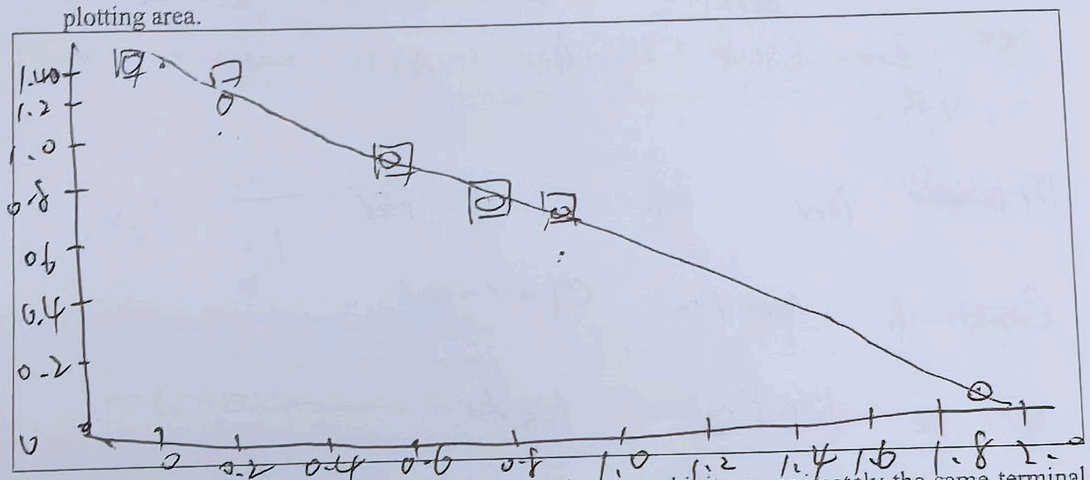
Table 4. Load Voltage, Current, and Power for the Original and Thevenin-Equivalent Circuits

R_L/Ω		Linear two-terminal circuit (Figure 10)			Thevenin Equivalent Circuit (Figure 11)			
Nominal	Measured	V_L /V	I_L /mA	P_L/mW	V_L /V	I_L /mA	P_L/mW	
0	0	0	/		0	/	/	
R_{TH} = 764	764.	~~0.657~~ 0.707	~~0.285~~ 0.91	0.6434	0.709	0.901	0.6388	
1k	1k.	0.731	0.72	0.528	0.735	0.72	0.5292	
2k	2.004k	0.938	0.45	0.422	0.945	0.45	0.4252	
5k	5.00k.	1.12	0.22	0.246	1.13	0.23	0.2599	
10k	~~9.96k~~	1.201	0.11	0.132	1.203	0.12	0.1564	
Open		1.406	~~0~~	0	1.403	0	0	

3. Load Voltage-Current Characteristic (8 pts)

Use the data in Table 4, plot a graph of V_L (on the vertical axis) versus I_L (on the horizontal axis).

- Use open circles for the original-network measurements. Use open squares for the Thevenin-equivalent measurements. Do not connect the measured data points.
- Plot the open-circuit point $(I_L, V_L) = (0, V_{OC})$ and the short-circuit point $(I_L, V_L) = (I_{SC}, 0)$, and connect them with a straight reference line.
- Include a descriptive title, axis labels with units, an appropriate scale, and a legend below the plotting area.



4. Do the original network and its Thevenin equivalent exhibit approximately the same terminal behavior? Support your answer with quantitative comparisons. (6 pts)

Yes

A Linear Line with Squares and circles in the same area

Experiment 3: Maximum Power Transfer

1. For each load resistance in Table 4, calculate the power delivered to the load.
2. Identify the load resistance that produces the largest measured power. (6 pts)

$P_{max,mas} = 0.707 \times 0.91 = 0.64337\,mW$

$0.707 \times 0.901 = 0.6368\,mW$

$P_L = \frac{V_{TH} \cdot R_L}{(R_{TH}+R_L)^2}$

$R_L = R_{TH}$, $1.4^2/(4\times760) = 0.645\,W$

Discussion Question

Design an experimental scheme to verify Norton's equivalent theorem for the circuit in Figure 2. Include parameter determination, construction of the Norton equivalent, measurement methods and so on. (8pts)

Determine I_N : Remove R_L and energize original network. Measure short-circuit current Set $I_N = I_{sc} = 1.84\,mA$ Also measure $V_{oc} = 1.406\,V$

Determine R_N the : $R_N = \frac{V_{oc}}{I_{sc}} = 764.13\,\Omega$

Construct Norton equivalent

Measure identical loads

Compare results

Finally, before leaving lab, turn off all equipment and return cables to their proper place. Leave your lab station clean and ready for other students to use. Thank you!